6.3) Problem Solving on Paper

A. Research the concept of “Information Gain Ratio” (which was discussed in lectures). In

100 words, discuss what is wrong with regular “Information Gain” and how does

“Information Gain Ratio” partially remedy this. Explain with one example.

B. Research the concept of cross-validation. In 100 words, discuss what is different between

StratifiedKFold and ShuffleSplit. Explain with one example.

Information Gain

Information Gain Ratio is the concept that is used to overcome and reduce bias towards multivalued attributes by including the number and size of branches when choosing the attribute. A tendency towards multivalued attributes may result in overfitting when creating a prediction model, thus ruining accuracy. Using the Information Gain Ratio saves the entropy/information gain from one state as it changes to another and calculates the information gain ratio after a split or change occurs. The information gain ratio is the information gain ratio to the intrinsic information.

Even using the information gain ratio may overcompensate and choose attributes with lower intrinsic information than other attributes due to overfitting. This can be fixed if only attributes with greater information gain are used.

This can be seen when making predictions and choosing the right amount of branches in a decision tree. Over or underfitting will return much less accurate, but machine learning models are predictions to the best of their ability.

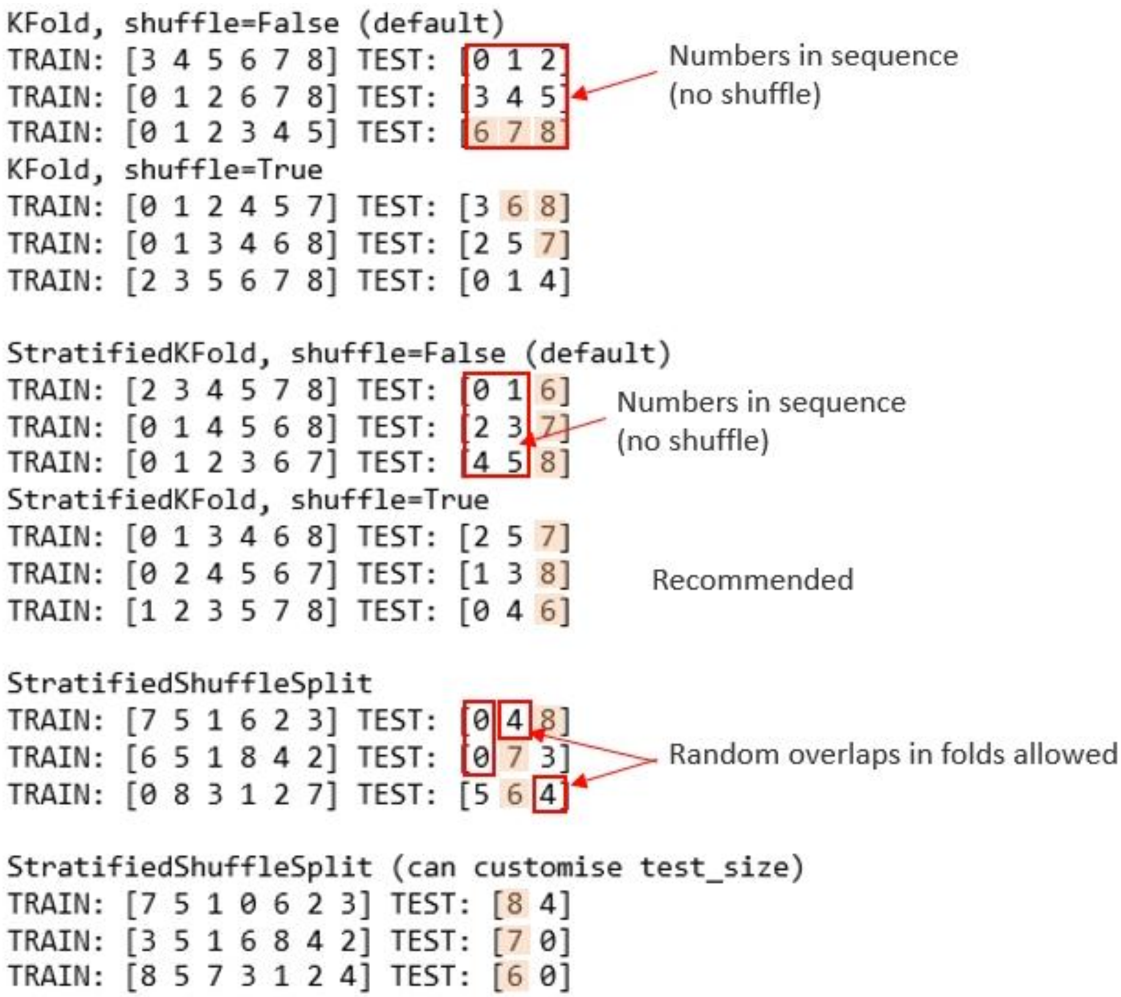
Cross-Validation

Cross-Validation is a method by which we train our model using subsets of the Dataset and then evaluate it using the leftover subset after k-1 for a 10 fold cross-validation; this is 9 training/testing subsets and one evaluating final subset. As a result, this allows for more accurate estimates with out of sample accuracy and more “efficient” use of data as every observation is used for both training and testing.

StratifiedKFold is a variation of KFold. StratifiedKFold first shuffles your data, then splits the data into n_splits parts and is finished. Now it will use each part as a test set**. It only and always shuffles data one time before splitting.**

StratifiedShuffleSplit is a variation of ShuffleSplit. It first shuffles your data, and then it also splits the data into n_splits parts. However, on top of this, StratifiedShuffleSplit picks one part to use as a test set. Then it repeats the same process n_splits - 1 other times to get n_splits - 1 different test set.

The difference is that StratifiedKFold shuffles and splits once at the beginning. Therefore the test sets do not overlap, while StratifiedShuffleSplit shuffles each time before splitting, and it splits n_splits times. Thus, the resulting test sets can overlap.



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